

Assignment No 20

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Q1.

```
import pandas as pd

df = pd.read_csv("ford_cleaned.csv")

X = df.drop("price", axis=1)
Y = df["price"]

print("Shape of X:", X.shape)
print("Shape of Y:", Y.shape)
```

```
... Shape of X: (17966, 8)
    Shape of Y: (17966,)
```

Q2.

Categorical Columns:

```
*** Index(['model', 'transmission', 'fuelType'], dtype='object')
```

First 5 Rows After One-Hot Encoding:

	year	mileage	tax	mpg	engineSize	model_ B-MAX	model_ C-MAX	\
0	2017	15944	150	57	1	0	0	
1	2018	9083	150	57	1	0	0	
2	2017	12456	150	57	1	0	0	
3	2019	10460	145	40	1	0	0	
4	2019	1482	145	48	1	0	0	

	model_ EcoSport	model_ Edge	model_ Escort	...	model_ Transit Tourneo	\
0	0	0	0	...	0	
1	0	0	0	...	0	
2	0	0	0	...	0	
3	0	0	0	...	0	
4	0	0	0	...	0	

	model_Focus	transmission_Automatic	transmission_Manual	\
0	0	1	0	
1	0	0	1	
2	0	0	1	
3	0	0	1	
4	0	1	0	

	transmission_Semi-Auto	fuelType_Diesel	fuelType_Electric	\
0	0	0	0	
1	0	0	0	
2	0	0	0	
3	0	0	0	
4	0	0	0	

```
2          0          0          0
3          0          0          0
4          0          0          0
```

```
      fuelType_Hybrid fuelType_Other fuelType_Petrol
0          0          0          1
1          0          0          1
2          0          0          1
3          0          0          1
4          0          0          1
```

```
[5 rows x 37 columns]
```

Q3.

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
df = pd.read_csv("ford_cleaned.csv")

num_cols = ['year', 'mileage', 'tax', 'mpg', 'engineSize']

scaler = StandardScaler()

df[num_cols] = scaler.fit_transform(df[num_cols])

print("First 5 Rows of Scaled Features:")
print(df[num_cols].head())
```

```
... First 5 Rows of Scaled Features:
   year  mileage    tax    mpg  engineSize
0  0.065128 -0.380998  0.591358 -0.020442  -0.811386
1  0.552866 -0.733359  0.591358 -0.020442  -0.811386
2  0.065128 -0.560132  0.591358 -0.020442  -0.811386
3  1.040605 -0.662640  0.510727 -1.738890   0.345070
4  1.040605 -1.123724  0.510727 -0.909294  -0.811386
```

Q4.

```
import pandas as pd
from sklearn.model_selection import train_test_split

df = pd.read_csv("ford_cleaned.csv")

X = df.drop("price", axis=1)
y = df["price"]

X = pd.get_dummies(X, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.33,
    random_state=42

    print("Shape of X_train:", X_train.shape)
    print("Shape of X_test :", X_test.shape)
    print("Shape of y_train:", y_train.shape)
    print("Shape of y_test :", y_test.shape)
```

```
... Shape of X_train: (12037, 34)
    Shape of X_test : (5929, 34)
    Shape of y_train: (12037,)
    Shape of y_test : (5929,)
```

Q5.

```
... Intercept:
-2098775.7025615196

Coefficients:

```

	Feature	Coefficient
0	year	1.047371e+03
1	mileage	-6.806506e-02
2	tax	-1.012340e+00
3	mpg	-8.101922e+01
4	engineSize	2.506197e+03
5	model_ C-MAX	1.098427e+03
6	model_ EcoSport	1.667128e+03
7	model_ Edge	9.563640e+03
8	model_ Escort	1.400243e+04
9	model_ Fiesta	1.179182e+03
10	model_ Focus	3.229532e+03
11	model_ Fusion	3.690363e+03
12	model_ Galaxy	6.596011e+03
13	model_ Grand C-MAX	1.496332e+03
14	model_ Grand Tourneo Connect	4.077367e+03
15	model_ KA	-2.331827e+02
16	model_ Ka+	-2.849109e+03
17	model_ Kuga	3.743432e+03
18	model_ Mondeo	2.992007e+03
19	model_ Mustang	1.223189e+04
20	model_ Puma	7.739733e+03

21	model_Ranger	2.501110e-12
22	model_S-MAX	6.401634e+03
23	model_Streetka	-9.094947e-12
24	model_Tourneo Connect	3.527424e+03
25	model_Tourneo Custom	6.724226e+03
26	model_Transit Tourneo	1.716558e+03
27	model_Focus	-1.435943e+01
28	transmission_Manual	-4.256625e+02
29	transmission_Semi-Auto	-1.280586e+02
30	fuelType_Electric	2.273737e-13
31	fuelType_Hybrid	8.157637e+03
32	fuelType_Other	8.966697e+02
33	fuelType_Petrol	-4.060427e+02

Q6.

▼ *** First 10 Predicted Values:
[6676.74792864 9175.10363172 9876.15109276 3808.41679251
2944.77931076 16623.10788237 27571.14532172 12128.40953688
11577.4693611 16774.32274023]

Comparison of Actual and Predicted Values:

	Actual Price	Predicted Price
0	6995	6676.747929
1	8999	9175.103632
2	7998	9876.151093
3	5491	3808.416793
4	3790	2944.779311
5	18200	16623.107882
6	22998	27571.145322
7	11000	12128.409537
8	7600	11577.469361
9	14985	16774.322740

Q7.

✓ ... R^2 Score: 0.8402100144860771

Q8.

▼ ... Model saved as: LR_ford_car.pkl
Scaler saved as: scaler.pkl
Columns saved as: columns.pkl
